

NIMIJITH NIMMI JAYAN

Graduate Software Engineer | AI & Python Developer | MSc Artificial Intelligence

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PROFESSIONAL SUMMARY

Graduate Software Engineer completing an MSc in Artificial Intelligence at Heriot-Watt University (September 2026), with practical experience building full-stack applications, REST API-driven systems, AI-integrated pipelines, and cloud-connected services. Proficient in Python, C#, Java, and SQL, with hands on use of Visual Studio Code, Git, Django, and Agile development workflows. Brings 13+ years of professional experience in digital operations, enterprise systems, and process-driven environments, providing a level of workplace maturity, accountability, and delivery discipline that sets this profile apart from a typical graduate candidate. Genuinely motivated by the intersection of software engineering and AI-driven automation, and eager to contribute to production grade development within a collaborative engineering team.

CORE TECHNICAL SKILLS

Languages:	Python, C#, Java, SQL
Software Engineering:	OOP, SDLC, Agile, Unit Testing, Debugging, Modular Architecture, Code Documentation, REST APIs
Backend & Frameworks:	Django, .NET WinForms, Authentication, Session Management, CRUD Operations, API Integration
AI & ML:	Machine Learning, Deep Learning, NLP, Conversational AI, RAG Pipelines, LSTM, SHAP, Reinforcement Learning, CNNs
Cloud & Integrations:	Azure Speech Services, Firebase, Deepgram API, WebRTC, Microsoft Azure (familiarity)
Data & Databases:	NumPy, Pandas, Scikit-learn, MongoDB, Neo4j, Cassandra, Redis, Data Pipelines
Dev Tools:	Visual Studio Code, Git, GitHub, Jupyter, IntelliJ, Linux, VS Code
Awareness:	Microsoft Copilot/AI Agent concepts, OpenAPI/REST, CI/CD principles, Containerisation, Automated Testing

KEY PROJECTS

- Multiparty AI-Human Spoken Dialogue System** *Python · Azure Speech · Deepgram · RAG · Firebase · WebRTC*
- Built a production-style conversational customer support system within an 8-person engineering team, integrating cloud speech APIs, a RAG based dialogue pipeline, real-time WebRTC audio handover, and Firebase signalling.
 - Contributed to system architecture decisions, evaluation design, ethics documentation, and cross-functional delivery in an Agile style workflow.
 - Analysed performance metrics including SUS scores, handover quality, and user satisfaction across 25 participants, demonstrating strong engineering evaluation and documentation skills.
- Hospital Management System** *Python · Django · REST · Authentication · Backend Development*
- Designed and built a full-stack web application with role-based access control for administrators, doctors, and staff, implementing modular backend logic for appointment scheduling, CRUD operations, and session management.
 - Applied software engineering best practices including separation of concerns, maintainable architecture, and structured testing. Available on GitHub.
- Mini Browser Application** *C# · .NET WinForms · OOP · Event-Driven Architecture*
- Developed a functional GUI based desktop application in C# implementing HTTP request handling, navigation history, bookmark management, and event-driven UI logic, directly relevant to the C#/.NET development environment.
- Explainable AI Financial Risk Forecasting System** *Python · LSTM · SHAP · PDDL · AI Planning (Ongoing Dissertation)*
- Designed a hybrid AI system combining LSTM-GARCH risk prediction, PDDL based AI planning, and SHAP explainability for dynamic hedging decisions, demonstrating strong applied AI and software problem-solving capability.
 - Completed requirements analysis, methodology design, ethics review, and structured evaluation planning.

Music Genre Classification *Python · CNN · Machine Learning · Model Evaluation*

- Built CNN and ML models for audio classification covering the full pipeline: preprocessing, augmentation, training, validation, testing, and performance evaluation, showcasing end-to-end software engineering and ML workflow.

AI Agents & Intelligent Systems *Java · A* Search · Reinforcement Learning · PDDL*

- Developed intelligent software agents using search algorithms, AI planning logic, and policy-based decision-making, directly aligned with Proclad's interest in AI-driven automation and Copilot agent development.

PROFESSIONAL EXPERIENCE

Customer Service & Digital Operations Lead | *India Post, Government of India* Jun 2011 - Jul 2024

- Managed digital adoption and rollout of enterprise systems across operational teams, contributing to improved transaction reliability across high-volume financial services environments (1,000+ customer interactions per month).
- Worked extensively with SAP-supported workflow systems for inventory, process coordination, and administrative operations, developing strong enterprise system proficiency and operational problem-solving skills.
- Cross-checked and validated savings, insurance, and financial transaction records ensuring accuracy and compliance in a high-volume, data-driven environment.
- Mentored and trained junior colleagues on digital tools, workflow procedures, and systems best practices, building strong communication, leadership, and knowledge transfer capability.
- Received formal commendation for outstanding contribution to technology adoption and digital process improvement.

Additional UK Employment:

Crew Member | McDonald's, Edinburgh | Sept 2025 - Present
Preload Sorter & Loader | UPS, Edinburgh | Nov 2024 - Sept 2025

EDUCATION

MSc Artificial Intelligence (Ongoing) | *Heriot-Watt University, Edinburgh* Sept 2024 - Sept 2026

- Relevant modules: Advanced Software Engineering, Software Engineering Foundations, Industrial Programming, Artificial Intelligence & Intelligent Agents, Data Mining & Machine Learning, Natural Language Processing, Big Data Management, Conversational Agents & Spoken Language Processing, Statistical Modelling & Analysis.
- Year 1 results (2024 to 2025): 7 A grades and 1 B across 8 modules. Top marks include Statistical Modelling and Analysis (95), Introduction to Interaction Design (90), AI and Intelligent Agents (89) and Introduction to Computer Systems (87).
- Year 2 Semester 1 provisional results (2025 to 2026): 4 A grades across all modules, including Industrial Programming (99), Biologically Inspired Computation (79), Natural Language Processing (75) and Data Mining and Machine Learning (73).

BSc Physics (2:1) | *Govt. Brennen College, Kannur University, India* Aug 2008 - Apr 2011

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Python Programming Certified User, Illinois Institute of Technology (via Entri Elevate), Oct 2025 | Grade: A+
- Python for Data Science and Machine Learning Essential Training, LinkedIn Learning (Ongoing)
- Learn JavaScript, Scrimba (Ongoing)
- Introduction to Java, LearnQuest, Coursera, May 2024

ADDITIONAL

- Volunteer, European Chatbot & Conversational AI Summit, March 2026
- Volunteer Retail Support, British Heart Foundation, Edinburgh, Jul to Sep 2025
- Languages: English (Fluent), Malayalam (Native), Hindi (Conversational)
- Right to work in the UK: Graduate Route Visa application planned December 2026 (no Tier 2 sponsorship required during the 2-year post-study period)

References available upon request.